Creativity Support Tools and Convergent Thinking: A Preliminary Review on Idea Evaluation and Selection

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Introduction

The **promotion of creativity** through technology has increased in recent years, and, as a result, the **use of CSTs has proliferated**.

Guilford [1] **divided creative thinking** into **divergent thinking** (ideation phase - generation of ideas) and **convergent thinking** (evaluation and selection of ideas).

In Banathy's "double diamond" [2], this classification was not only further explored but also linked, leading to many theories about the underlying nature of creative processes. **Our specific interest** is in the **dichotomy between evaluating and selecting ideas**.

Some cases in the **history of HCI demonstrate** that people tend to **reject good ideas**, which can lead to **weaker solutions** to problem-solving [3].

We carried out a **preliminary review on the CSTs that support convergent thinking** to **obtain insights** that can **guide us** in the **future development** of CSTs.

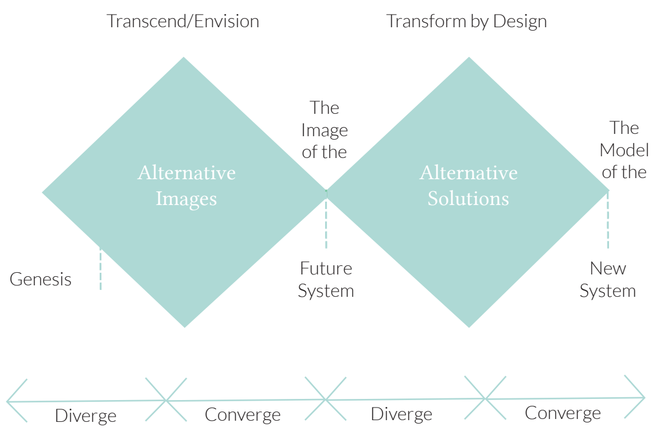

Figure 1: Adaptation of the Double Diamond model to our research.

Method

The **objective** of this first phase was to **analyze and evaluate** the **direction the CSTs focused on convergent thinking are taking**.

Our **tool selection method** was based on the **CST definition given by Frich et al. [4]**.

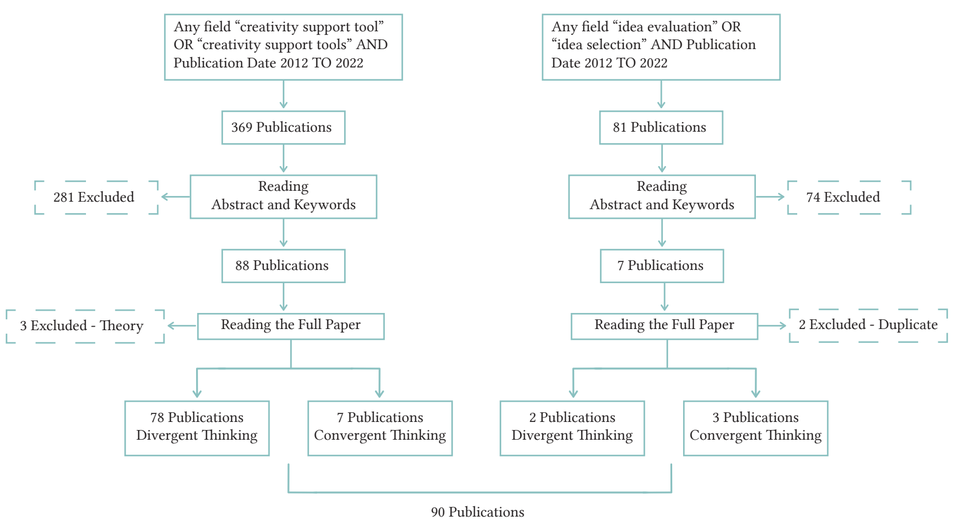

Figure 2: A descriptive flowchart of the search process.

We obtained a **sample of 90 CSTs**, with **80 articles** focusing on **divergent thinking** and **10** on **convergent thinking**.

The **stages** of convergent thinking **were classified** using two methodologies **(Human vs. Machine)**. Idea **evaluation** was **mainly machine supported**, while idea **evaluation** was **only human-supported**.

Results

The research covered the **domains** of **Affective Computing**, **Visual Art & Design**, **Sound Production**, and **Undefined**.

The **Web Application** is the **most developed system** to support convergent thinking, with **50%** of the articles.

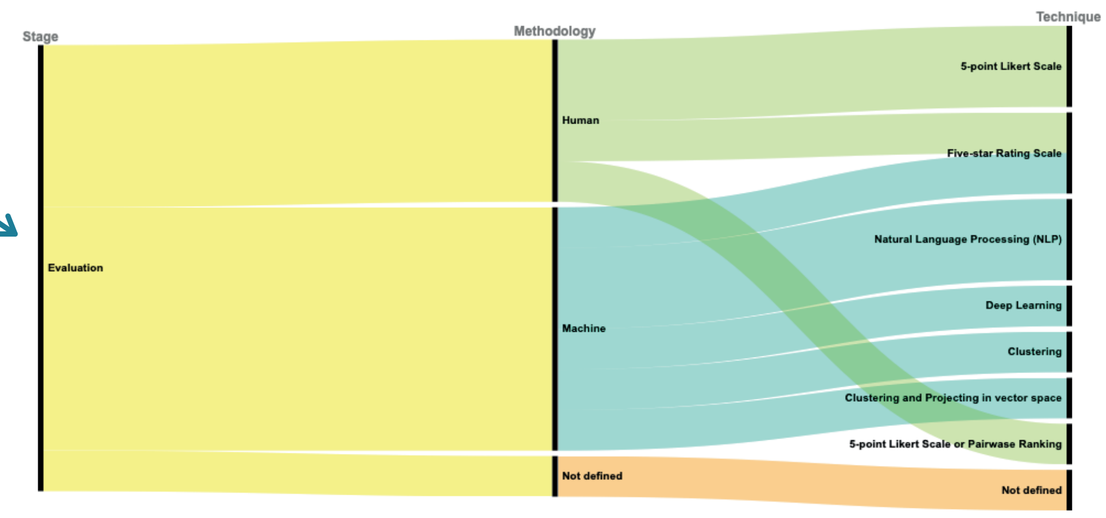

Figure 3: Grouping of the ten publications through the stage they investigated.

We believe the **idea selection step** could **benefit** from updating **AI methods**, e.g., Computer Vision methods (analyzing images, sketches, and videos).

Conclusions

Most CSTs support divergent thinking, which **overloads the range of ideas**, and the **main issue is choosing the best idea**.

Responding to our initially proposed objective, we present **6 guidelines** that can **enhance the functionality** of CSTs and **increase integration in the users' workflow**.

Numeration	CST Guidelines	Goal
G1	**New domains: Media & Communication and Health** These domains were not included in the sample but could improve by introducing these tools.	Expand domains and thus contribute to an increase in CSTs that support convergent thinking.
G2	**Application of Human-Centered Artificial Intelligence (HCAI)** Use AI methods based on Deep Learning [23] to capture human cognitive abilities, creating a hybrid human-machine system.	Highlight the acquisition, understanding, and creation of mental connections to obtain the best idea (s) and improve human performance through reliable, safe, and trustworthy systems.
G3	**Combining AI techniques with Information Visualization methods** The idea evaluation and selection steps can take advantage of classification, grouping, filtering, and recommendation techniques and present the information to the user through appropriate information visualization methods.	Support the balance between human-machine decision-making, taking advantage of sensemaking, and the amount of information managed in the creative context.
G4	**Use of Ubiquitous Technology** CSTs for mobile devices, such as smartphones or tablets, are tools with a high degree of usability.	Insert in the context of the user's use a type of system that he is already familiar with. As they are ubiquitous systems, they are already part of the users' workflow, it does not require the introduction of an extra system.
G5	**Changing the type of environments - Support for groups and specialists** Beneficial to take advantage of the previous knowledge of the users, being able to take advantage of this information and thus develop CSTs with more functionalities, becoming more complete.	Monetize meetings, brainstorming sessions, and workshops so that CSTs can support novice and expert users.
G6	**Features Combination** Divergent thinking along with convergent thinking.	Supporting the entire creative process and reducing information overload.

At the moment, a central issue in the area is **cooperation between humans and machines**.

We hope that **applying these guidelines will result** in **better and more human-centered CSTs** and that our synthesis could facilitate further research on the topic.

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